

MEGHA JOSHI

Experienced statistician with strong quantitative background in causal inference methods. Successfully led multiple quantitative projects demonstrating proficiency in devising analytic strategies and communicating results to stakeholders.

EDUCATION

- 2021 • **The University of Texas at Austin**
PhD in Quantitative Methods  Austin, TX
- 2014 • **Bryn Mawr College**
BA in Art History and Psychology  Bryn Mawr, PA

WORK EXPERIENCE

- 2025
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Present • **Lead Data Analyst**
Progressive Insurance  Remote
 - Develop guidelines and templates to design and implement experimental and non-experimental designs like randomized AB tests with and without clustering, difference-in-differences, synthetic control, and observational designs to accurately infer causal effects of new campaigns and strategies on customer experiences.
 - Research new methodologies to handle advanced experimental designs, like Group Sequential Testing. Conduct Monte Carlo simulation studies to evaluate performance of different testing methods, ensuring accuracy of lift estimates.
 - Consult on implementing statistical analyses in Python and R. Advise on statistical topics such as multiple testing, power analysis, multilevel data analysis, frequentist versus Bayesian inference.
 - Develop methodology to forecast results of AB tests to guide business decisions.
- 2021
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2025 • **Senior Quantitative Researcher**
American Institutes for Research  Remote
 - Directed multi-year projects evaluating educational programs and policies. Broke down projects into actionable tasks. Delegated the analytic tasks and mentored junior staff to drive research effectiveness.
 - Developed predictive models using supervised learning techniques, including linear and logistic regression, generalized boosted models and Bayesian Additive Regression Trees, to forecast outcomes.
 - Led quantitative impact analyses by developing analytic strategies for projects using causal inference and machine learning.
 - Procured contracts worth over \$3M by leading research proposals and helping devise analytic strategies for new projects.
 - Produced reports with intuitive visualizations of impact results for presentation to technical as well as non-technical audiences. Automated report generation in R, reducing reporting time by 50%. Created dashboards presenting the results using R Shiny.

CONTACT INFO

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 meghapsimatrix.com
 github.com/meghapsimatrix
 469-235-3003
 Austin, Texas
For more information, please
contact me via email.

SKILLS

Statistical Software: R, Python,
SQL, SAS

Version Control: Git

Project Management: Asana, Trello

RESEARCH INTERESTS

Causal inference

Meta-analysis

Machine learning

R PACKAGES

[wildmeta](#)

[simhelpers](#)

[metaselection](#)

*This resume was made with the R
package [pagedown](#).*

2021

Data Scientist

Analyst Institute

📍 Remote

- Developed optimal codebase to conduct inferential analysis on data with over a hundred million rows and data with complex structures, ensuring accuracy and reducing analysis time by 40%.
- Designed the methodological and analytical strategy to conduct the inferential analysis. Solved methods related issues.

2016

2021

Graduate Research Assistant

The University of Texas at Austin

📍 Austin, TX

- Led the methods team for a project examining the effects of teacher preparation programs on teacher retention in Texas.
- Evaluated the impact of a college preparatory program using propensity score analysis with generalized boosted modeling.
- Evaluated the impact of teacher preparation programs on teacher retention using survival analysis.
- Developed and implemented the analytical strategy.
- Produced reports and presentations detailing the results to be presented to a non-technical audience.



TEACHING EXPERIENCE

2015

2021

Graduate Teaching Assistant

The University of Texas at Austin

📍 Austin, TX

- Assisted in the following courses: Causal Inference; Data Analysis, Simulation and Programming in R; Research Design; Survey of Multivariate Methods; Fundamental Statistics; and Statistics in Market Analysis.
- Led weekly problem-solving sessions through office hours; effectively communicated complex statistical methods to students; and, fostered interest in methodological research.



SELECTED PUBLICATIONS

2022

Cluster wild bootstrapping to handle dependent effect sizes in meta-analysis with a small number of studies

Research Synthesis Methods

Joshi, M., Pustejovsky, J. E., & Beretvas, S. N.

2019

The performance of multivariate methods for two-group comparisons with small samples and incomplete data

Multivariate Behavioral Research, Pages 1-18

Pituch, K. A., Joshi, M., Cain, M. E., Whittaker, T. A., Chang, W., Park, R., & McDougall, G. J.

2019

Evaluating the Transition to College Mathematics course in Texas high schools: Findings from the first year of implementation

Greater Texas Foundation

Pustejovsky, J. E., & Joshi, M.